

ECG Arrhythmia CNN Report

Wajih

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Introduction

This report details the development of a Convolutional Neural Network (CNN) for ECG arrhythmia detection using the MIT-BIH Arrhythmia Database, achieving a balanced accuracy of 0.88 on records 100, 101, and 203.

1 Methodology

1.1 Data Collection and Preprocessing

Records 100, 101, and 203 were used, each with 40,000 samples, totaling 120,000 samples. The `ECGDataLoader` class handles loading and preprocessing:

- **load_ecg_data:** Loads signals from CSV files and annotations from .atr files, concatenating signals and adjusting R-peak offsets.
- **segment_beats:** Segments signals into 256-sample windows around R-peaks, filtering out invalid segments and assigning binary labels (0 for normal, 1 for arrhythmic).
- **preprocess_data:** Normalizes segments to [0, 1] and reshapes for CNN input.

Here's the core of `segment_beats`:

```

1 def segment_beats(self, signal, r_peaks, labels):
2     half_window = self.window_size // 2
3     segments = []
4     segment_labels = []
5     for i in range(len(r_peaks)):
6         r_peak = r_peaks[i]
7         start = r_peak - half_window
8         end = r_peak + half_window
9         if 0 <= start and end < len(signal):
10            segment = signal[start:end]
11            segments.append(segment)
12            segment_labels.append(1 if labels[i] == 'A' else 0)
13    return np.array(segments), np.array(segment_labels)

```

This method ensures segments align with labels, addressing earlier mismatches (e.g., 20 vs. 66 samples).

1.2 Model Architecture and Training

The `ECGCNNModel` class defines a CNN with convolutional layers, max-pooling, dropout, and dense layers, compiled with binary cross-entropy loss. The training process in `main.py` splits data into 70% training, 15% validation, and 15% test sets, with stratification to maintain class balance:

```

1 X_train, X_temp, y_train, y_temp = train_test_split(
2     segments, segment_labels, test_size=0.3, random_state=42,
3     stratify=segment_labels
4 )

```

```

4 stratify_second = y_temp if all(count >= 2 for count in counts_temp)
   else None
5 X_val, X_test, y_val, y_test = train_test_split(
6     X_temp, y_temp, test_size=0.5, random_state=42, stratify=
   stratify_second
7 )

```

The dynamic stratification avoids errors when a class has fewer than 2 samples (e.g., only 1 arrhythmic beat in `y_temp`).

1.3 Visualization

The `ECGPlotter` class plots ECG segments and confusion matrices, aiding in visual validation of model predictions.

2 Results

Metric	Value
Accuracy	0.97
Precision	0.88
Recall	0.88
F1 Score	0.88

Table 1: Final Model Performance

Class distribution improved from 264:2 to 424:51 with record 203, enabling effective learning of arrhythmic patterns.

3 Challenges and Solutions

- **Imbalance:** Initially, only 2 arrhythmic beats were present; adding record 203 increased this to 51.
- **Split Mismatch:** Resolved by adjusting stratification logic to use subset labels (`y_temp`).
- **Learning Minority Class:** The CNN learned arrhythmic patterns after increasing minority class samples.

4 Conclusion and Future Work

The model achieved a balanced accuracy of 0.88, exceeding the 0.85 target. Future enhancements could include class weights, adding records 207 and 208, or deeper architectures.